Mathematics Analysis and Approaches HL

The exploration of Fourier Series and Transform to Investigate the Analysis of Signals

Candidate Number: lfg856

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Introduction and Aim:

The aim of this exploration is to explore how Fourier transformations can be applied to manipulate signals – such as audio, communication, and image compression – and demonstrate the benefits of using complex numbers for analysis. Specifically, this will investigate audio processing by analyzing a sound wave to identify its frequency components. Many topics from the AA HL curriculum will be used in this exploration, such as integration, Euler's form of complex numbers, trigonometric functions, and converging/diverging sequences. First, the transformations, analysis (decomposition), and synthesis (construction) will be introduced. This will then be followed by performing the Fourier Transform on a square wave and a randomized wave mathematically and computationally.

I chose this topic for my exploration, because I have a passion for Electrical Engineering. In the past I have built communication protocols utilizing lasers which still have a delay in processing due to the on/off feature. Thus, by investigating Fourier Transformations I will be able to build my own communication without delay with frequencies. Furthermore, I have always found the idea of imaginary numbers paradoxical in real world analysis as they have no physical meaning.

Completeness of Orthogonal Functions:

The foundation of Fourier Transform is the ability to express any function as the weighted sum of cosines and sines. To understand this, I will introduce the concept of orthogonal functions. The collection of functions $S = \{\phi_1, \phi_2, \phi_3, \cdots, \phi_n, \cdots\}$ is orthogonal on $[a, b]$ if:

$(\phi_i, \phi_j) = \int_a^b \phi_i \phi_j \, dx = \begin{cases} 0 \; if \; i \neq j \\ c > 0 \; if \; i = j \end{cases}$ for any pair i and j, where (f, g) denotes the inner product of two functions f and g (Gabardo)

Let the function $f(x)$ be defined as:

$$f(x) = c_1\phi_1 + c_2\phi_2 + \cdots + c_n\phi_n + \cdots = \sum_{i=1}^{k} c_i\phi_i$$

For every $0 \leq n$, we can write the following:

$$(f, \phi_n) = \int_a^b f(x)\phi_n \, dx = \int_a^b \left(\sum_{i=1}^{k} c_i \phi_i\right)\phi_n \, dx = \sum_{i=1}^{k} \int_a^b c_i \phi_i \phi_n \, dx \ldots (1)$$

Because S is orthogonal, the integral inside the summation is evaluated to be 0 for any i other than n. Thus, (1) simplifies into:

$$c_n \int_a^b (\phi_n)^2 \, dx = (f, \phi_n)$$

$$c_n = \frac{(f, \phi_n)}{\int_a^b (\phi_n)^2 \, dx} \ldots (2)$$

From (2), every coefficient of $f(x)$ can be found, showing that every function can be expressed as the sum of the collection of orthogonal functions. However, this does not hold for every function. For example, consider the function $f(x)$ which is orthogonal to every function in the collection, and not equal to 0. Evaluating (2) shows that every coefficient is 0, making $f(x) = 0$, contradicting the claim that $f(x)$ is non-zero.

An orthogonal set is called complete if the fact that $f(x)$ is orthogonal to every function implies that $f(x)$ is orthogonal to itself. From this definition I claim that $S_1 = \{1, \cos(x), \sin(x), \cos(2x), \sin(2x), \ldots\}$ is a complete orthogonal set on $[-\pi, \pi]$. Therefore, I have to prove the following conditions have to be held:

$$\int_{-\pi}^{\pi} cos(nx)\, cos(mx) \, dx = 0 \ldots (3a)$$

$$\int_{-\pi}^{\pi} sin(nx)\, sin(mx) \, dx = 0 \ldots (3b)$$

$$\int_{-\pi}^{\pi} cos(nx)\, sin(mx) \, dx = 0 \ldots (3a)$$

To prove these conditions, the product of trigonometric functions is expressed as the sum, as listed below (‘3.4: Sum-To-Product’).

$$For\ 3a:\ cos(\alpha)\,cos(\beta) = \frac{1}{2}(cos(\alpha - \beta) + cos(\alpha + \beta))$$

$$For\ 3b:\ sin(\alpha)\,cos(\beta) = \frac{1}{2}(sin(\alpha - \beta) + sin(\alpha + \beta))$$

$$For\ 3c:\ sin(\alpha)\,sin(\beta) = \frac{1}{2}(cos(\alpha - \beta) - cos(\alpha + \beta))$$

Below is the proof for (3a) with a similar proof for (3b) and (3c).

$$\int_{-\pi}^{\pi} cos(nx)\,cos(mx)\,dx = \int_{-\pi}^{\pi} \frac{cos((n-m)x)}{2}dx + \int_{-\pi}^{\pi} \frac{cos((n+m)x)}{2}dx = \frac{1}{2}\left[\frac{\sin((n-m)x)}{n-m}\right]_{-\pi}^{\pi} + \frac{1}{2}\left[\frac{\sin((n+m)x)}{n+m}\right]_{-\pi}^{\pi}$$

This equation is evaluated to 0 as desired. Furthermore, these functions satisfy the equality statement (when i=j):

$$(1,1) = \int_{-\pi}^{\pi} (1)\,dx = 2\pi$$

$$(co\,s(nx)\,,co\,s(nx)) = \int_{-\pi}^{\pi} (\cos^2(nx))\,dx = \pi$$

$$(si\,n(mx)\,,si\,n(mx)) = \int_{-\pi}^{\pi} (\sin^2(mx))\,dx = \pi$$

The proof of completeness of this set is outside the scope of the paper and is assumed to be true. Therefore, any function, $f(x)$, can be expressed in the form of sines and cosines.

$$f(x) = a_0 + \sum_{n=1}^{N} (a_n \cos(nx) + b_n \sin(nx)) \ldots (4)$$

This function is the Fourier Series, which is discussed further below, with the coefficients being defined.

a_0 arises for the case of when $n = 0$

I also claim that $S_2 = \{\ldots, e^{-nix}, e^{-(n-1)ix}, \ldots, 1, \ldots, e^{(n-1)ix}, e^{nix}, \ldots\}$ to be a complete orthogonal set. However, because the elements in this set are complex, the inner product is redefined as the following to preserve the properties of the inner product (Osgood):

$$(\phi_i, \phi_j) = \int_a^b \phi_i \overline{\phi_j}\, dx = \begin{cases} 0 \; if \; i \neq j \\ c > 0 \; if \; i = j \end{cases}$$

Thus, similar to the above we have to prove the following conditions, where e^{-bix} is the complex conjugate of e^{bix}:

$$\int_{-\pi}^{\pi} e^{aix}\, dx = 0 \ldots (5a)$$

$$\int_{-\pi}^{\pi} e^{aix} e^{-bix}\, dx = 0 \ldots (5b)$$

To prove (5a) and (5b), the following identity is used, which is proved by expanding the exponentials with Euler's Identity, $e^{ix} = \cos(x) + i\sin(x)$ (Gregersen):

$$i\sin(x) = \frac{e^{ix} - e^{-ix}}{2} \ldots (6)$$

Below is the proof for (5b) which is a generalized version of (5a):

$$\int_{-\pi}^{\pi} e^{(a-b)ix}\, dx = \left[\frac{e^{(a-b)ix}}{(a-b)i}\right]_{-\pi}^{\pi} = \frac{e^{(a-b)i\pi} - e^{-(a-b)i\pi}}{(a-b)i} = \frac{2isi\,n\big((a-b)\pi\big)}{(a-b)i} = \frac{2si\,n\big((a-b)\pi\big)}{(a-b)}$$

Because a and b are integers, the sin in the numerator simplifies to 0 as desired. Additionally, the equality statement of these functions is satisfied (when i=j):

$$(1,1) = \int_{-\pi}^{\pi} (1)\, dx = 2\pi$$

$$\left(e^{aix}, e^{aix}\right) = \int_{-\pi}^{\pi} e^{aix} e^{-aix}\, dx = \int_{-\pi}^{\pi} (1)\, dx = 2\pi$$

Therefore, any function, $f(x)$, can be expressed in the form of complex exponentials.

$$f(x) = \sum_{n=-N}^{N} c_n e^{njx}\, , wherej = \sqrt{-1} \ldots (7)$$

Additionally, it can be proven that both s_1 and s_2 work over any period P, however, the proof of this is outside the scope of this paper. For the rest of the paper, it is assumed that this fact that any period P can be used. Furthermore, this paper assumes that (4) and (7) will converge even for non-continuous functions; the proof of this lies outside the scope of this paper.

Derivation of coefficients in the Fourier Series:

To extend (4) into P, we rewrite (4) into the following equation:

$$f(x) = a_0 + \sum_{n=1}^{N} (a_n \cos(wnx) + b_n \sin(wnx))\, , \text{where } w = \frac{2\pi}{P} \ldots (8)$$

We first find the coefficients of a_i. Multiply both sides by $\cos(wmx)$, then integrate both sides over P.

$$f(x)\cos(wmx) = a_0 \cos(wmx) + \sum_{n=1}^{N} (a_n \cos(wnx) \cos(wmx)) + \sum_{n=1}^{N} (b_n \sin(wnx) cos\ (wmx))$$

$$\int_P f(x)\, cos(wmx) dx = \int_P a_0 \cos(wmx) dx + \int_P \sum_{n=1}^{N} (a_n \cos(wnx) \cos(wmx)) dx +$$

$$\int_P \sum_{n=1}^{N} (b_n \sin(wnx) cos\ (wmx))\, dx \ldots (9)$$

Above we have already proved that the integral of sin multiplied by cos evaluates to 0. For positive integer values a, the function $\cos(awt)$ has exactly a complete oscillations inside the interval P. Because it has an integer number of complete oscillations the area both under and over the curve sum to 0.

Therefore, we can simplify (9) to the following, by assuming that $m > 0$:

$$\int_P f(x)\, cos(wmx)dx = \int_P \sum_{n=1}^{N} (a_n \cos(wnx) \cos(wmx))dx \ldots (10)$$

$$\int_P f(x)\, cos(wmx)dx = \sum_{n=1}^{N} a_n \int_P \frac{1}{2} (cos\big((n+m)wx\big)dx + \cos\big((n-m)wx\big))dx$$

$$\int_P f(x)\, cos(wmx)dx = \frac{1}{2} \sum_{n=1}^{N} a_n \left(\int_P \cos\big((n+m)wx\big)\, dx + \int_P \cos\big((n-m)wx\big)\, dx \right)$$

$$\int_P f(x)\, cos(wmx)dx = \frac{1}{2} \sum_{n=1}^{N} a_n \left(\int_P \cos\big((n-m)wx\big)\, dx \right) \ldots (11)$$

We see that $\int_P \cos\big((n-m)wx\big)\, dx$ also completes $|n-m|$ oscillations, so it is 0 except when $m = n$:

$$\int_P \cos\big((n-m)wx\big)\, dx = \begin{cases} 0\ if\ n \neq m \\ \int_P 1dx = P\ if\ n = m \end{cases}$$

Therefore (9) can simplify into:

$$\int_P f(x)\, cos(wmx)dx = \frac{a_m}{2}(P)$$

$$a_m = \frac{2}{P} \int_P f(x)\, cos(wmx)dx$$

To complete this proof, we consider the case when $m = 0$. From (10):

$$\int_P f(x)\, cos(0)dx = \int_P \sum_{n=0}^{N} (a_n \cos(wnx) \cos(0))dx$$

$$\int_P f(x)dx = \sum_{n=0}^{N} a_n \int_P cos(wnx)dx = a_0 \int_P cos(0)dx + \sum_{n=1}^{N} a_n \int_P cos(wnx)dx \ldots (12)$$

The summation evaluates to 0, because the integral inside the summation becomes 0 as it is a cosine function evaluated over its period. Therefore (12) becomes:

$$\int_P f(x)dx = Pa_0$$

$$a_0 = \frac{\int_P f(x)dx}{P}$$

Similarly, the coefficients of b_i can be found the same way. To review the coefficients of a Fourier Series are as follows:

$$a_0 = \frac{\int_P f(x)dx}{P} \ldots (13a)$$

$$a_n = \frac{2}{P}\int_P f(x)\, cos(wnx)dx \ldots (13b)$$

$$b_n = \frac{2}{P}\int_P f(x)\sin(wnx)\, dx \ldots (13c)$$

Note that (11a) is also given as the average value for the function of $f(x)$, and that w is the frequency of $f(x)$. Additionally, the Dirichlet conditions guarantee that the Fourier Series converges. These conditions are outside the scope of this paper and are assumed to be true (Baraniuk).

<u>Equivalence Between Trigonometric and Exponential Series:</u>

To allow for easier manipulation as well as simpler expression, (7) is often used to express the Fourier Series instead of (4). Similar to how we extended (4) to P, we do the same thing for (7):

$$f(x) = \sum_{n=-N}^{N} c_n e^{nwjx} \ldots (14)$$

From here I claim that (8) is equivalent to (14). To start, we consider the constant terms a_0 and c_0. Because both (8) and (14) only have one, these constant terms must be equal. Therefore, $a_0 = c_0$. For the rest of the terms, consider the parts of both function which oscillates the same number of times during P.

$$a_n \cos(wnx) + b_n \sin(wnx) = c_{-n} e^{-nwjx} + c_n e^{nwjx} \dots (15)$$

From Euler's Formula, we know that a complex exponential can be written as the sum of a cos and sin arguments. Additionally, because the left side is real, the right side has to be real. Since e^{-nwjx} is the complex conjugate of e^{nwjx}, c_{-n} must be the complex conjugate of c_n in order for the imaginary terms to cancel. Setting $c_n = c_{n,r} + jc_{n,i}$, leads $c_{-n} = c_{n,r} - jc_{n,i}$. Note that c_n is a complex coefficient, but a_n and b_n aren't because we introduced an imaginary term into S_2. In order for the function to collapse into real components, the coefficients have to be complex to produce real numbers. Therefore (15) simplifies into the following,

$$(c_{n,r} - jc_{n,i})(\cos(-nwx) + \mathrm{j}\sin(-nwx)) + (c_{n,r} + jc_{n,i})(\cos(nwx) + \mathrm{j}\sin(nwx))$$

$$= (c_{n,r} - jc_{n,i})(\cos(nwx) - \mathrm{j}\sin(nwx)) + (c_{n,r} + jc_{n,i})(\cos(nwx) + \mathrm{j}\sin(nwx))$$

$$= 2c_{n,r}\cos(nwx) - 2c_{n,i}\sin(nwx) + j\left(\left(c_{n,i} - c_{n,i}\right)\cos(nwx) + \left(c_{n,r} - c_{n,r}\right)\sin(nwx)\right)$$

$$= 2c_{n,r}\cos(nwx) - 2c_{n,i}\sin(nwx) \dots (16)$$

Equation the trigonometric functions in (15) and (16) we get the following:

$$a_n = 2c_{n,r}$$

$$b_n = -2c_{n,i}$$

Therefore,

$$c_n = \frac{a_n}{2} - j\frac{b_n}{2}$$

From (13b) and (13c),

$$c_n = \frac{1}{2}(a_n - jb_n) = \frac{1}{2}\left(\frac{2}{P}\int_P f(x)\,cos(wnx)dx - j\frac{2}{P}\int_P f(x)\,sin(wnx)dx\right)$$

$$= \frac{1}{P}(\int_P f(x)(\cos(wnx) - j\,sin(wnx))dx = \frac{1}{P}(\int_P f(x)(\cos(-wnx) + j\,sin(-wnx))dx$$

$$= \frac{1}{P}(\int_P f(x)(e^{-nwjx})dx) \ldots (17)$$

Therefore, the trigonometric expression of the Fourier Transform can also be written in the exponential form. Also note that the above proof utilized e^{-nwjx} which seemingly has a negative frequency of $-w$, which doesn't make sense for sinusoids; but the physical nature of signals allows for this.

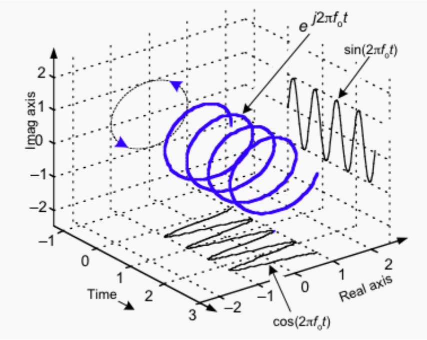

Figure 1: Nature of how a signal is typically seen, it is constructed with two orthogonal sinusoidal functions (Lyons)

In Figure 1, signals that are rotating to the right have a positive frequency, whereas those rotating to the left have a negative frequency. This can be thought of as going forward and backward in time, respectively.

Time-Domain and Frequency Domain:

Up to this point, the behavior of functions has only been evaluated on a time domain. However, in practice this is rarely used as very little information can be drawn from these graphs. Instead, frequency domains are often used to model behavior. Assume that we want to graph the function $A\cos(\omega t + \varphi)$ in the frequency domain. This function has a phase or shift of φ, frequency of ω, and amplitude A. From here we can uniquely determine the graph in the frequency domain as an impulse at ω of height A.

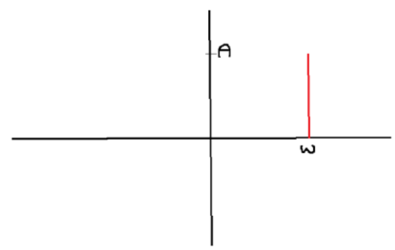

Figure 2: A graphical representation of $A\cos(\omega t + \varphi)$ in the frequency domain

However, in the frequency domain given by Figure 2, the phase is not accounted for. To account for the phase, the following manipulation is performed ('Shifts'):

$$FT\big(f(t - t_0)\big) = e^{-j\gamma t_0}X(\gamma)$$

γ and $X(\gamma)$ are defined as frequency and frequency domain function and are expanded on in the next section. Additionally the phase shift can also be found through the Fourier Transform stated below.

Extending Into an Aperiodic Function:

So far, only periodic functions have been considered; however, many functions are not periodic. Thus, to move from periodic functions to aperiodic functions, we simply let $P \to \infty$. When this happens, the coefficients get closer together as w becomes smaller, allowing nw to take up any value as w is very small, whereas n can be any integer value. From (17), we define the continuous function $X(\gamma)$, where $\gamma = nw$.

$$Pc_n = (\int_P f(x)(e^{-nwjx})dx) \; as \; P \to \infty$$

$$X(\gamma) = (\int_P f(x)(e^{-\gamma jx})dx) \ldots (18)$$

(18) is known as the analysis equation and is how the frequency domain of a function is derived. (18) often outputs a complex number, which has the amplitude and phase encoded in it. Let $X(\gamma) = a + bj$. The amplitude is given by the magnitude, $\sqrt{a^2 + b^2}$, and the phase is given by the angle, $arctan\left(\frac{b}{a}\right)$ (Gubner). Because the real and imaginary parts never interact due to the nature of complex number arithmetic, the magnitude never interferes with the phase and vice versa.

Likewise, we can find the synthesis equation which builds the time domain of a function from the frequency domain. Performing some manipulations on (14) and substituting in (18) we get the following equations. Furthermore, $w = \Delta\gamma$ because the difference between each set of coefficients is the first harmonic frequency, w.

$$f(x) = \sum_{n=-\infty}^{\infty} Pc_n e^{nwjx}\left(\frac{1}{P}\right)$$

$$f(x) = \sum_{n=-\infty}^{\infty} X(\gamma)e^{\gamma jx}\left(\frac{1}{\left(\frac{2\pi}{w}\right)}\right)$$

$$f(x) = \frac{1}{2\pi}\sum_{n=-\infty}^{\infty} X(\gamma)e^{\gamma jx}(\Delta\gamma)$$

$$f(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(\gamma)e^{nwjx}d\gamma \ldots (19)$$

(18) and (19) are collectively known as the Fourier Transform, as a function in the frequency domain is able to be transformed into a function of the time domain and vice versa.

Applying Knowledge to Analyze Functions:

Example 1: Rectangular Pulse Function:

Consider an even pulse function that has a period of 2, an amplitude of 1, and a pulse width of 1. Therefore, the function is defined as follows, where mod denotes the modulo function.

$$f(x) = \begin{cases} 1 \text{ if x mod } 2 \in (-0.5,0.5) \\ 0 \text{ otherwise} \end{cases}$$

Because this is an even function, the Fourier Series does not contain any sin terms. Therefore, (8) becomes:

$$f(x) = a_0 + \sum_{n=1}^{N} (a_n \cos(wnx)) = \sum_{n=0}^{N} (a_n \cos(wnx))$$

Solving for the coefficients with (13a) and (13b) we find that:

$$a_0 = \frac{(1)(1)}{2} \ldots (20)$$

$$a_n = \frac{2}{(2)} \int_P (1)\, cos(\pi n x) dx = \left[\frac{\sin(\pi n x)}{\pi n}\right]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{\sin(n\pi)}{n\pi} \ldots (21)$$

(20) is found from finding the area of one period which is simply the area of the rectangular pulse found by multiplying the pulse width by the amplitude. (21) is by integrating only over the pulse width, as anywhere else in the period it is 0, making the integral 0. Therefore, we find that

$$f(x) = \frac{1}{2} + \frac{\sin(\pi)}{\pi}\cos(\pi x) + \frac{\sin(2\pi)}{2\pi}\cos(2\pi x) + \cdots + \frac{\sin(i\pi)}{i\pi}\cos(i\pi x) + \cdots$$

Plotting the first nine values of this function we see that as more terms are a part of the Fourier Series, the more accurate the graph becomes in the time domain.

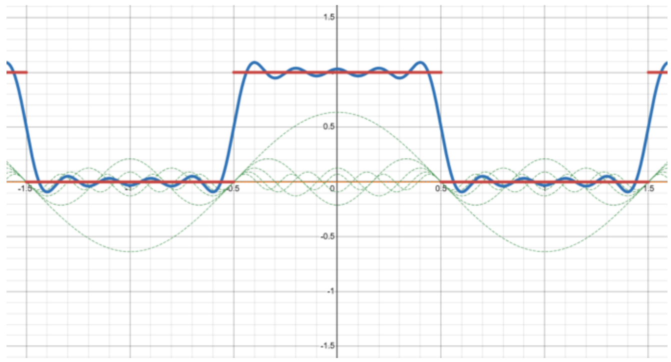

Figure 2: The dark red line denotes the rectangular pulse function, and the dark blue lines denotes the Fourier Series. The dotted green and orange lines are the individual sinusoidal functions that form the Fourier Series, with green denoting every odd n and orange denoting the even n. Note that the rectangular pulse function graphed is not a true pulse function because it lacks the vertical lines at the discontinuities but is used for simplicity. The functions that are graphed are shown in Appendix A ('Desmos')

The overshoot in oscillatory behavior around the jump discontinuities (-1.5, -0.5, 0.5, 1.5) can be explained by Gibb's phenomenon. This states that there will always be an overshoot near the discontinuity, which is around 9% (Baraniuk). As more terms are added, the ripples remain, but the width of them approaches zero. Thus, the reconstruction is exactly the original except for the points of discontinuity. Since Dirichlet Conditions guarantees a finite number of discontinuities, the pulse functions exhibits a non-uniform convergence.

From the Fourier Transform, we are also able to find the frequency domain graph of $f(x)$. From the same logic stated above regarding taking the integral of $f(x)$ over 0.5 to -0.5, we have the following from (18):

$$X(\gamma) = (\int_{P} f(x)(e^{-\gamma j x})dx) = \int_{-\frac{1}{2}}^{\frac{1}{2}} (1)e^{-\gamma j x}dx = \left[\frac{e^{-\gamma j x}}{-\gamma j}\right]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{1}{\gamma j}(e^{-\frac{\gamma j}{2}} - e^{\frac{\gamma j}{2}}) \ldots (22)$$

Therefore, (22) becomes:

$$X(\gamma) = \frac{2j\sin\left(\frac{\gamma}{2}\right)}{\gamma j} = \frac{2}{\gamma}\sin\left(\frac{\gamma}{2}\right) \ldots (23)$$

Graphing the amplitudes of the frequency domain we have the following:

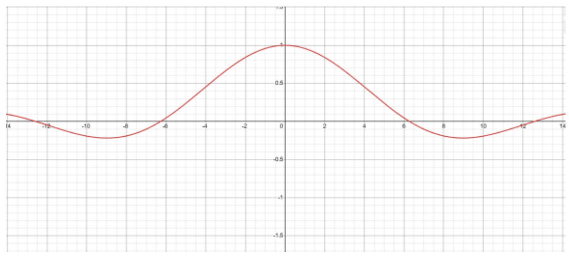

Figure 3: Graph of the amplitude of $X(\gamma)$ with respect to the frequency γ ('Desmos')

Note the resemblance between (21) and (23). Because only the cosine terms are present in an even function, the amplitude is equivalent to the coefficient a_n. Graphing these impulses on the frequency domain gives the same trace as (23).

For completeness it will be shown that the same result follows, if instead we use the complex exponential form, as stated in (17), and applying the identities and manipulations listed above:

$$c_n = \frac{1}{2}\int_{-\frac{1}{2}}^{\frac{1}{2}} (1)e^{-n\pi jx}\,dx = \frac{1}{2}\left[\frac{e^{-n\pi jx}}{-\gamma\pi j}\right]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{\sin(n\pi)}{n\pi}\ as\ desired$$

Example 2: Random Periodic Continuous Function:

Consider the following function:

$$f(x) = e^{-3|x|}\sin\big(2(x+1)\big), where\ x\epsilon[-\pi,\pi]$$

From (13a), (13b), (13c), we can derive the following Fourier Series:

$$a_0 = \frac{\int_{-\pi}^{\pi} e^{-3|x|}\sin\big(2(x+1)\big)\,dx}{2\pi} = \frac{1}{2\pi}\left(\int_{-\pi}^{0} e^{3x}\sin\big(2(x+1)\big)\,dx + \int_{0}^{\pi} e^{-3x}\sin\big(2(x+1)\big)\,dx\right)$$

$$a_0 = \frac{1}{13\pi}(3\sin(2) - 3e^{-3\pi}\sin(2)) \ldots (24)$$

$$a_n = \frac{2}{2\pi}\int_P f(x)\, cos(nx) dx = \frac{1}{\pi}\left(\int_{-\pi}^{\pi} e^{-3|x|}\sin\big(2(x+1)\big)\cos(nx)\, dx\right)$$

$$a_n = \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} e^{-3|x|}(\sin\big((2-n)x+2\big) + sin\ ((2+n)x+2)\, dx\right)$$

$$a_n = \left(\frac{6e^{-3\pi}(n^2+13)(e^{3\pi}\sin(2) - si\,n(2+n\pi))}{(n^4+10n^2+169)\pi}\right) \ldots (25)$$

$$b_n = \frac{2}{2\pi}\int_P f(x)\sin(nx)\, dx = \frac{1}{\pi}\left(\int_{-\pi}^{\pi} e^{-3|x|}\sin\big(2(x+1)\big)\sin(nx)\, dx\right)$$

$$b_n = \frac{1}{2\pi}\left(\int_{-\pi}^{\pi} e^{-3|x|}\cos\big((2-n)x+2\big)\, dx - \int_{-\pi}^{\pi} e^{-3|x|}\cos\big((2+n)x+2\big)\, dx\right)$$

$$b_n = \left(\frac{e^{-3\pi}(4n)(3e^{3\pi}\cos(2) - 6\sin(n\pi+2)}{(n^4+10n^2+169)\pi}\right) \ldots (26)$$

Substituting (24), (25), (26), we can derive the following with (5):

$$f(x) = a_0 + (a_1\cos(x) + b_1\sin(x)) + (a_2\cos(2x) + b_2\sin(2x)) + \cdots + (a_i\cos(ix) + b_n\sin(ix)) + \cdots$$

Plotting the first 11 values of this series we see the following function:

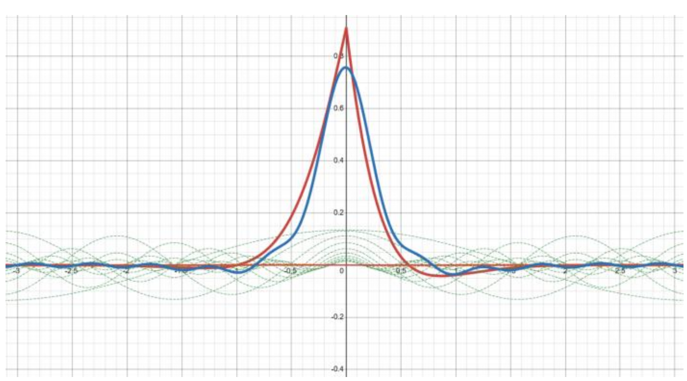

Figure 4: The dark red line denotes $f(x)$, and the dark blue lines denotes the Fourier Series. The dotted green and orange lines are the individual sinusoidal functions that form the Fourier Series, with green denoting every cosine associated term and orange denoting the even sine associated term. The functions that are graphed are shown in Appendix B ('Desmos')

Similar to the previous example, we can find the frequency domain of $f(x)$ with (18).

$$X(\gamma) = (\int_{P} (e^{-\gamma jx})dx) = \int_{-\pi}^{\pi} e^{-3|x|} \sin\big(2(x+1)\big)\, e^{-\gamma jx} dx$$

$$= \int_{-\pi}^{0} e^{3x-\gamma jx} \sin\big(2(x+1)\big)\, dx + \int_{0}^{\pi} e^{-3x-\gamma jx} \sin\big(2(x+1)\big)\, dx$$

Evaluating these integrals using integration by parts yields:

$$X(\gamma) = \frac{2e^{-3\pi} \sin(2)\, (3e^{3\pi}(\gamma^2+13) + \gamma(\gamma^2+5)\sin(\pi\gamma) - 3(\gamma^2+13)cos\,(\pi\gamma)}{\gamma^4 + 10\gamma^2 + 169}$$

$$+ j\frac{4e^{-3\pi} \cos(2)\, ((\gamma^2-13)\sin(\pi\gamma) + 6e^{3\pi}\gamma - 6\gamma co\,s(\pi\gamma)}{\gamma^4 + 10\gamma^2 + 169}$$

Plotting the amplitude of the frequency domain we have the following:

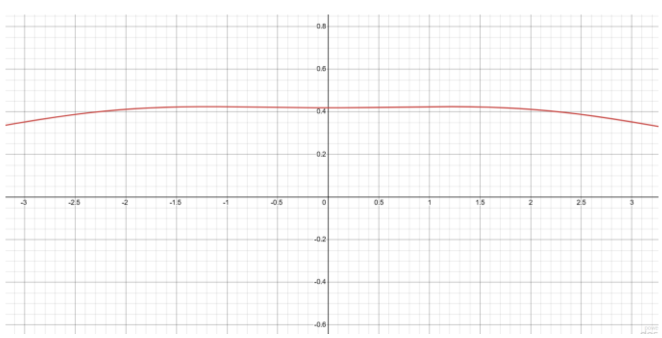

Figure 5: Graph of the amplitude of $X(\gamma)$ with respect to the frequency γ ('Desmos')

Note the shape of this graph. The amplitude at n = 1 and n = 2, were relatively the same, but for larger n, this magnitude started to decrease at a faster rate, as exemplified by the graph.

Conclusion:

To reiterate, the aim of this paper was to explore the methodology of using Fourier Series and Transforms to decompose signals into understandable sinusoidal functions, and to understand the impact of complex numbers in the matter. The usage of complex numbers to more easily represent the Fourier Series and enable the Fourier transform proves its usefulness. Both the Fourier Series and Transform were applied to practical examples of functions and noise reduction to verify both methodologies. However, I found Fourier Transforms to be more useful in not only finding the time-based domain but also extracting information to be used inside the frequency domain, demonstrating the necessity of complex numbers. Because of their definition, it is impossible to mix the real and imaginary parts of the complex number, allowing for easier calculations, while balancing two different values.

While this exploration explains how orthogonal sets allow for functions to be expressed in the sum of exponentials and sinusoidal functions, the Dirichlet Conditions are assumed to be true, without any explanation. These conditions lie at the center of Fourier Series and Transforms and are the reasons why orthogonal sets work. Thus, proving this condition, would enable are better understanding of Fourier Transforms. Additionally, another limitation is that I don't look into Discrete Fourier Transforms and Fast Fourier Transforms, which are prominently used in real-world applications over the Fourier Transform discussed in this paper. This is because of the uncertainty principle of Fourier Series, which states that the more complex a signal is, you can not have perfect precision in both the time and frequency domain simultaneously. Improving this limitation would allow me to see how the majority of signal processing is conducted in the real world.

This exploration may be extended by considering Laplace Transformations, which have a greater practical application, as they filter out the negative frequencies. Additionally, it may be of interest to look into using Fourier Transforms to solve for solutions in partial differential equations.

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Appendix:

Appendix A: Graph of the Fourier Series for Example 1:

Step Function: $\frac{((-1)^{floor(x+0.5)}+1)}{2}$

Fourier Series: $\mathrm{a}_0 + f_1(x) + f_2(x) + f_3(x) + f_4(x) + f_5(x) + f_6(x) + f_7(x) + f_8(x) + f_9(x)$

$$\mathrm{a}_0 = \frac{1}{2}$$

$$f_1(x) = \frac{2}{\pi}\sin\left(\frac{\pi}{2}\right)\cos(\pi x)$$

$$f_2(x) = \frac{2}{2\pi}\sin\left(\frac{2\pi}{2}\right)\cos(2\pi x)$$

$$f_3(x) = \frac{2}{3\pi}\sin\left(\frac{3\pi}{2}\right)\cos(3\pi x)$$

$$f_4(x) = \frac{2}{4\pi}\sin\left(\frac{4\pi}{2}\right)\cos(4\pi x)$$

$$f_5(x) = \frac{2}{5\pi}\sin\left(\frac{5\pi}{2}\right)\cos(5\pi x)$$

$$f_6(x) = \frac{2}{6\pi}\sin\left(\frac{6\pi}{2}\right)\cos(6\pi x)$$

$$f_7(x) = \frac{2}{7\pi}\sin\left(\frac{7\pi}{2}\right)\cos(7\pi x)$$

$$f_8(x) = \frac{2}{8\pi}\sin\left(\frac{8\pi}{2}\right)\cos(8\pi x)$$

$$f_9(x) = \frac{2}{9\pi}\sin\left(\frac{9\pi}{2}\right)\cos(9\pi x)$$

Appendix B: Graph of the Fourier Series for Example 2:

Function: $e^{-3abs(x)}\sin\big(2(x+1)\big)$

Fourier Series: $a_0 + \big(f_1(x) + g_1(x)\big) + \big(f_2(x) + g_2(x)\big) + \big(f_3(x) + g_3(x)\big) + \big(f_4(x) + g_4(x)\big) + \big(f_5(x) + g_5(x)\big) + \big(f_6(x) + g_6(x)\big) + \big(f_7(x) + g_7(x)\big) + \big(f_8(x) + g_8(x)\big) + \big(f_9(x) + g_9(x)\big) + \big(f_{10}(x) + g_{10}(x)\big) + \big(f_{11}(x) + g_{11}(x)\big)$

$$a_0 = \frac{1}{13\pi}(3\sin(2) - 3e^{-3\pi}\sin(2))$$

$$f_1(x) = \frac{6e^{-3\pi}(1^2+13)(e^{3\pi}\sin(2) - \sin(2+(1)\pi))}{(1^4+10(1)^2+169)\pi}\cos(x)$$

$$g_1(x) = \frac{e^{-3\pi}\big(4(1)\big)(e^{3\pi}\cos(2) - \sin(2+(1)\pi))}{(1^4+10(1)^2+169)\pi}\sin(x)$$

$$f_2(x) = \frac{6e^{-3\pi}(2^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (2)\pi))}{(2^4 + 10(2)^2 + 169)\pi}\cos(2x)$$

$$g_2(x) = \frac{e^{-3\pi}\big(4(2)\big)(e^{3\pi}\cos(2) - \sin(2 + (2)\pi))}{(2^4 + 10(2)^2 + 169)\pi}\sin(2x)$$

$$f_3(x) = \frac{6e^{-3\pi}(3^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (3)\pi))}{(3^4 + 10(3)^2 + 169)\pi}\cos(3x)$$

$$g_3(x) = \frac{e^{-3\pi}\big(4(3)\big)(e^{3\pi}\cos(2) - \sin(2 + (3)\pi))}{(3^4 + 10(3)^2 + 169)\pi}\sin(3x)$$

$$f_4(x) = \frac{6e^{-3\pi}(4^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (4)\pi))}{(4^4 + 10(4)^2 + 169)\pi}\cos(4x)$$

$$g_4(x) = \frac{e^{-3\pi}\big(4(4)\big)(e^{3\pi}\cos(2) - \sin(2 + (4)\pi))}{(4^4 + 10(4)^2 + 169)\pi}\sin(4x)$$

$$f_5(x) = \frac{6e^{-3\pi}(5^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (5)\pi))}{(5^4 + 10(5)^2 + 169)\pi}\cos(5x)$$

$$g_5(x) = \frac{e^{-3\pi}\big(4(5)\big)(e^{3\pi}\cos(2) - \sin(2 + (5)\pi))}{(5^4 + 10(5)^2 + 169)\pi}\sin(5x)$$

$$f_6(x) = \frac{6e^{-3\pi}(6^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (6)\pi))}{(6^4 + 10(6)^2 + 169)\pi}\cos(6x)$$

$$g_6(x) = \frac{e^{-3\pi}\big(4(6)\big)(e^{3\pi}\cos(2) - \sin(2 + (6)\pi))}{(6^4 + 10(6)^2 + 169)\pi}\sin(6x)$$

$$f_7(x) = \frac{6e^{-3\pi}(7^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (7)\pi))}{(7^4 + 10(7)^2 + 169)\pi}\cos(7x)$$

$$g_7(x) = \frac{e^{-3\pi}\big(4(7)\big)(e^{3\pi}\cos(2) - \sin(2 + (7)\pi))}{(7^4 + 10(7)^2 + 169)\pi}\sin(7x)$$

$$f_8(x) = \frac{6e^{-3\pi}(8^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (8)\pi))}{(8^4 + 10(8)^2 + 169)\pi}\cos(8x)$$

$$g_8(x) = \frac{e^{-3\pi}\big(4(8)\big)(e^{3\pi}\cos(2) - \sin(2 + (8)\pi))}{(8^4 + 10(8)^2 + 169)\pi}\sin(8x)$$

$$f_9(x) = \frac{6e^{-3\pi}(9^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (9)\pi))}{(9^4 + 10(9)^2 + 169)\pi}\cos(9x)$$

$$g_9(x) = \frac{e^{-3\pi}\big(4(9)\big)(e^{3\pi}\cos(2) - \sin(2 + (9)\pi))}{(9^4 + 10(9)^2 + 169)\pi}\sin(9x)$$

$$f_{10}(x) = \frac{6e^{-3\pi}(10^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (10)\pi))}{(10^4 + 10(10)^2 + 169)\pi}\cos(10x)$$

$$g_{10}(x) = \frac{e^{-3\pi}\big(4(10)\big)(e^{3\pi}\cos(2) - \sin(2 + (10)\pi))}{(10^4 + 10(10)^2 + 169)\pi}\sin(10x)$$

$$f_{11}(x) = \frac{6e^{-3\pi}(11^2 + 13)(e^{3\pi}\sin(2) - \sin(2 + (11)\pi))}{(11^4 + 10(11)^2 + 169)\pi}\cos(11x)$$

$$g_{11}(x) = \frac{e^{-3\pi}\big(4(11)\big)(e^{3\pi}\cos(2) - \sin(2 + (11)\pi))}{(11^4 + 10(11)^2 + 169)\pi}\sin(11x)$$